

Abstract

Since carbon dioxide and air pollutants are co-emitted, achieving deep decarbonization is expected to result in significant air quality co-benefits. However, some decarbonization pathways will yield greater air quality benefits than others, and measures such as biomass combustion and carbon capture and sequestration can increase downwind air pollution.

We hypothesize that considering air quality benefits explicitly when developing decarbonization strategies has the potential to result in achieving air quality benefits sooner and to increase the overall magnitude of benefits.

To test this hypothesis, the air quality impacts of three scenarios are evaluated: a net zero scenario (NetZero) and two alternatives (Benefits-Lo and Benefits-Hi) that achieve equivalent decarbonization but endogenize the health costs associated with air pollution.

The scenarios are evaluated using EPA's GLIMPSE air-climate-energy planning tool, which is built upon the Global Change Analysis Model. For each scenario, GCAM is allowed to identify a cost-effective pathway for achieving the net zero target.

We identify how these pathways differ with respect to technology choices and assess the health benefits associated with each.

Introduction

Problem:

- Decarbonization is expected to yield air quality co-benefits
- These benefits may differ based on the decarbonization pathway

Hypothesis:

- By endogenizing health benefits directly into the GLIMPSE framework, we can identify alternative pathways with much greater air quality co-benefits

Overview of approach:

- Models
 - GLIMPSEv1.1 is used to simulate decarbonization pathways and track changes in co-emitted air pollutants (epa.gov/glimpse)
 - COBRAv5.1 is used to quantify air quality and health outcomes (epa.gov/cobra)
- Scenarios:
 - NetZero – National CO₂ emissions are constrained to follow a decreasing linear trajectory through 2050. This trajectory was used within the Energy Modeling Forum 37 exercise.
 - Benefits-Lo and Benefits-Hi – net zero scenarios that endogenize health impacts to **NO_x**, **SO₂**, and **PM_{2.5}** equivalent to low and high estimates of health damages.

Results

Monetized health benefits (a):

- Health benefits through 2050 are shown for three scenarios.
 - NetZero minimizes control costs
 - Benefits-Lo and Benefits-Hi endogenize estimates of air quality damages
- Both alternative scenarios yield substantially more benefits than NetZero.
 - Benefits-Lo: 2.5x more
 - Benefits-Hi: 3x more
- A discount rate of 2% was used in COBRA to calculate these values.

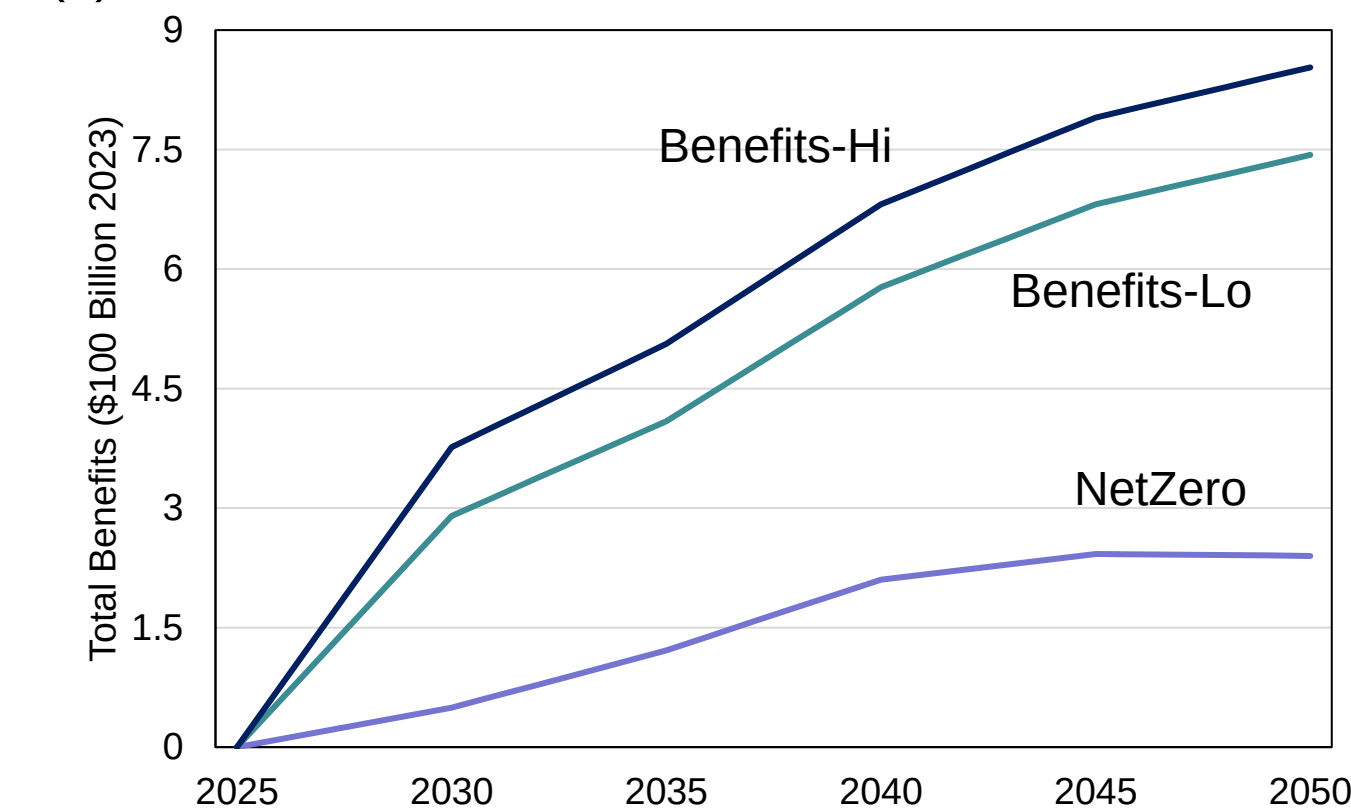
Sectoral contribution to benefits (b):

- The bulk of benefits can be attributed to emission changes within three categories:
 - industrial sector (75%)
 - industrial processes (9%)
 - residential buildings (14%)

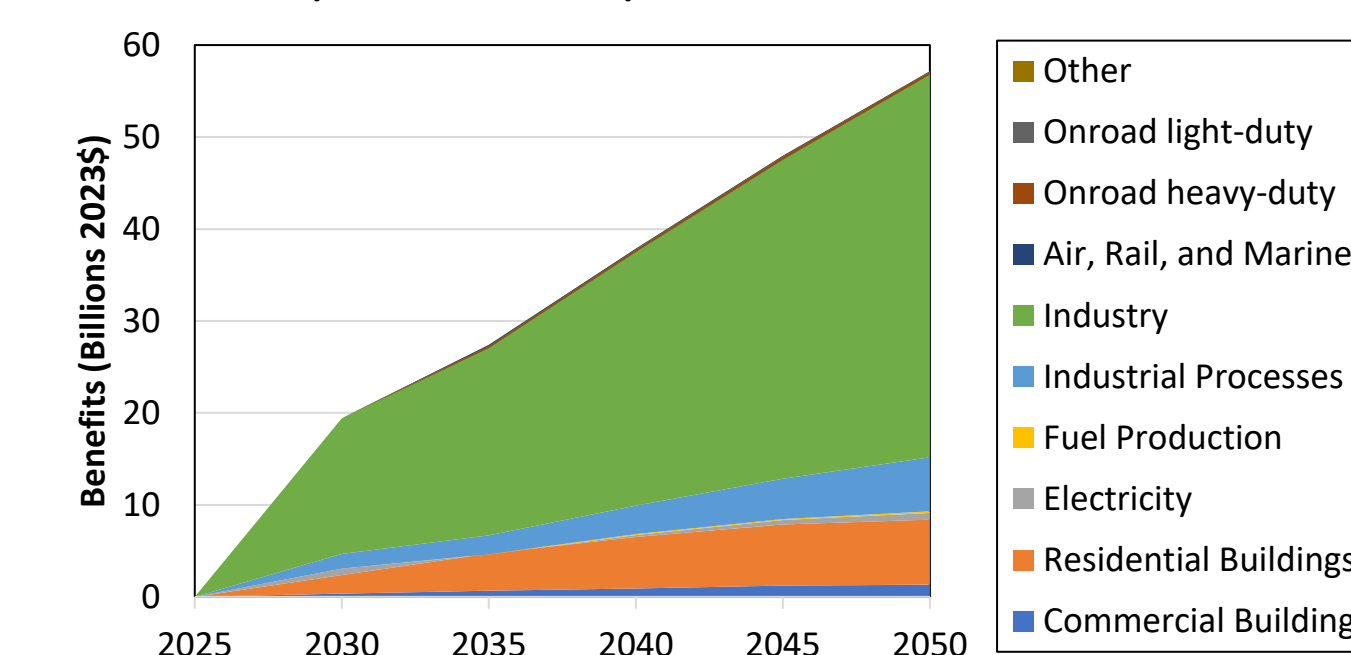
Sectoral responses for High-Estimate (c,d,e):

- Sectoral fuel use changes also provide information about the model's response.
- Between Benefits-Hi and NetZero:
 - Industrial energy use (c) decreased, and there was a transition to electricity from other fuels
 - Electrification in buildings (d) also increased, offsetting biomass and gas
 - In Transportation (e), liquid fuels increased, and electricity decreased
 - reflects additional use of biofuels and higher electricity prices as electricity demand from other sectors grows.

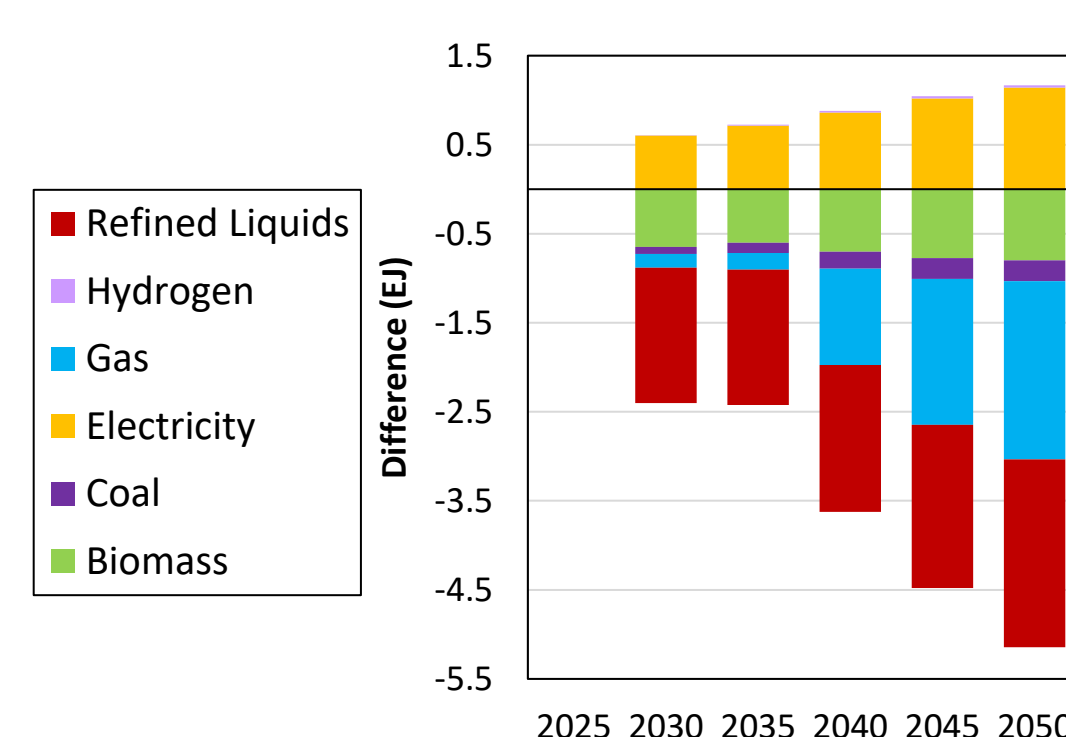
(a) Monetized Benefits of three Net Zero Scenarios



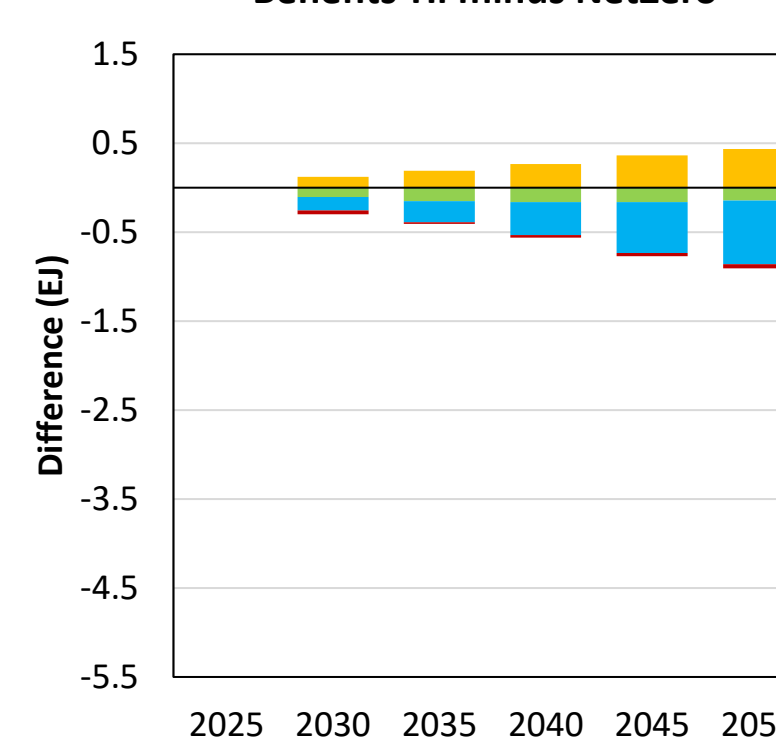
(b) Difference in Benefits of Emissions Reductions by Sector (PM_{2.5}, NO_x, SO₂): Benefits-Hi minus NetZero



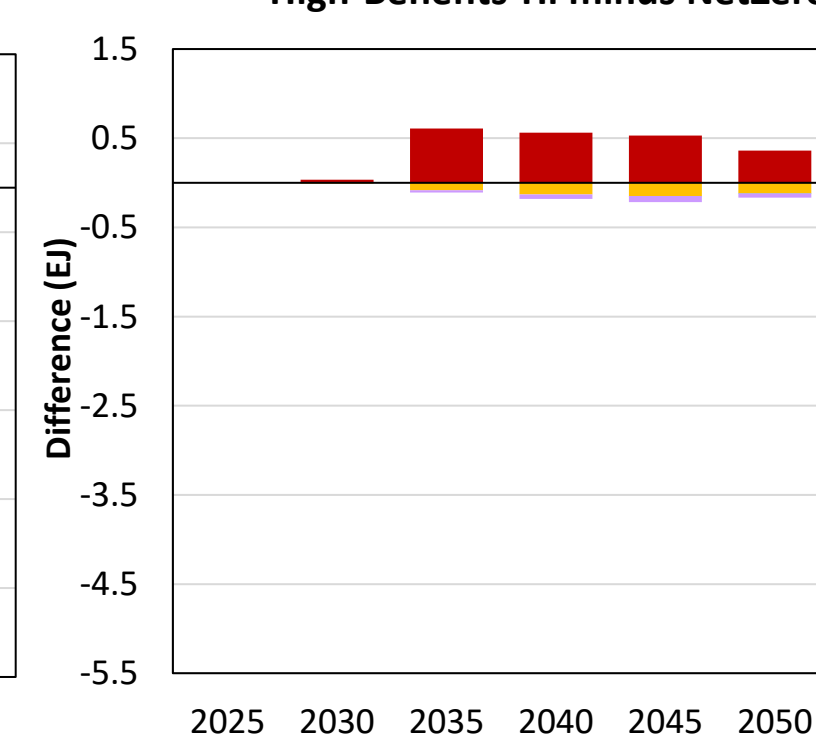
(c) Energy Consumption in Industry: Benefits-Hi minus NetZero



(d) Energy Consumption in Buildings: Benefits-Hi minus NetZero



(e) Energy Consumption in Transportation: High-Benefits-Hi minus NetZero



Methodology

- Simulate Reference scenario in GLIMPSE
- Develop COBRA input files for Reference scenario
- Develop alternative COBRA input files by decreasing Reference levels of NO_x, SO₂, and PM_{2.5} by 10%
- Run COBRA for Reference and each alternative (2030-2050)
- Calculate low and high national \$/ton benefit estimates for NO_x, SO₂, and PM_{2.5} from COBRA results
- Construct NetZero and two variants that endogenize \$/ton benefits: Benefits-Lo and Benefits-Hi
- Simulate NetZero, Benefits-Lo, and Benefits-Hi in GLIMPSE and COBRA
- Investigate whether and how endogenizing benefits changes results affects the NetZero strategy

Conclusions

Results:

- By endogenizing health impacts, we successfully identified Net Zero decarbonization pathways that greatly increased health benefits.
 - Under the Benefits-Hi scenario, benefits were 3x greater
- The effects of endogenizing benefits are primarily in GCAM's industrial sector.
- Major responses included:
 - decreased use of refined liquids, natural gas, coal, and biomass in industry
 - reduced use of coal in electricity production, with increases in renewables, particularly wind and solar (not shown)

Limitations:

- Industrial sector has comparatively less detail than other sectors
- Inherent uncertainty in predicting future processes and outcomes
- National benefit/ton estimates ignore sectoral and regional differences

Next Steps:

- Repeat using new GCAM detailed industrial sector when available
- Develop and apply health factors that differentiate by type of source and by region of the country